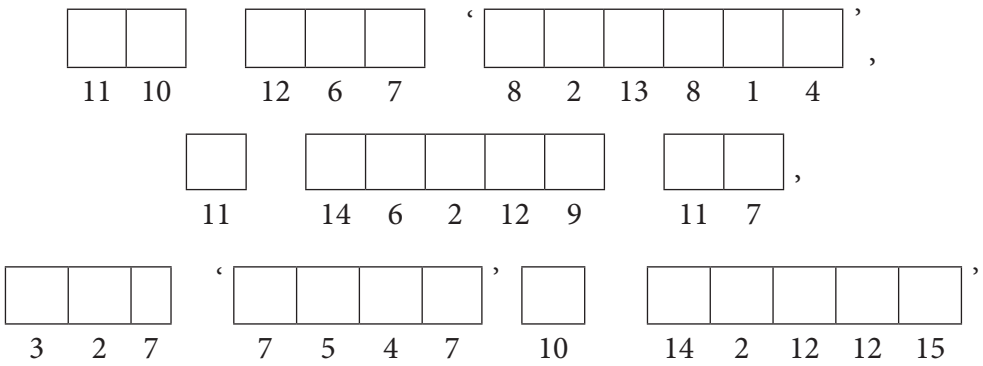
WORKSHEET

Chapter 8: Normal distribution – Worded problems 1

The most exciting phrase to hear in science, the one that heralds new discoveries …



Isaac Asimov – science fiction author

Questions

1 The Year 12 female population at a local school has a mean height of 163 cm, with a standard deviation of 5 cm. Use the normal distribution to determine the probability that a female student, selected at random, is:

a taller than 170 cm

b shorter than 160 cm

c between 160 and 170 cm

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d between 165 and 168 cm

e taller than 155 cm.

2 A crop of genetically modified tomato seedlings has a mean growth of 7 cm in the first 2 weeks, with a standard deviation of 1.5 cm. If a seedling is chosen at random, what is the probability, using the normal distribution, that it is:

a taller than 8.2 cm?

b shorter than 5.5 cm?

c between 6.6 cm and 7.9 cm?

d shorter than 9 cm?

e between 5 and 6 cm?

3 The mean first-year salary of a group of university graduates is $29 300, with a standard deviation of $2000. If one of those graduates is selected at random, what is the probability that the graduate earns:

a more than $30 000?

b between $31 000 and $33 000?

c less than $24 000?

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d less than \$34 000?

e between \$28 000 and \$30 000?

Solutions

A 0.1859

B 0.6449

D 0.9082

E 0.3321

F 0.9906

H 0.9452

I 0.3632

K 0.0808

N 0.1655

O 0.2119

R 0.004

S 0.1596

T 0.1587

U 0.2743

Y 0.379

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